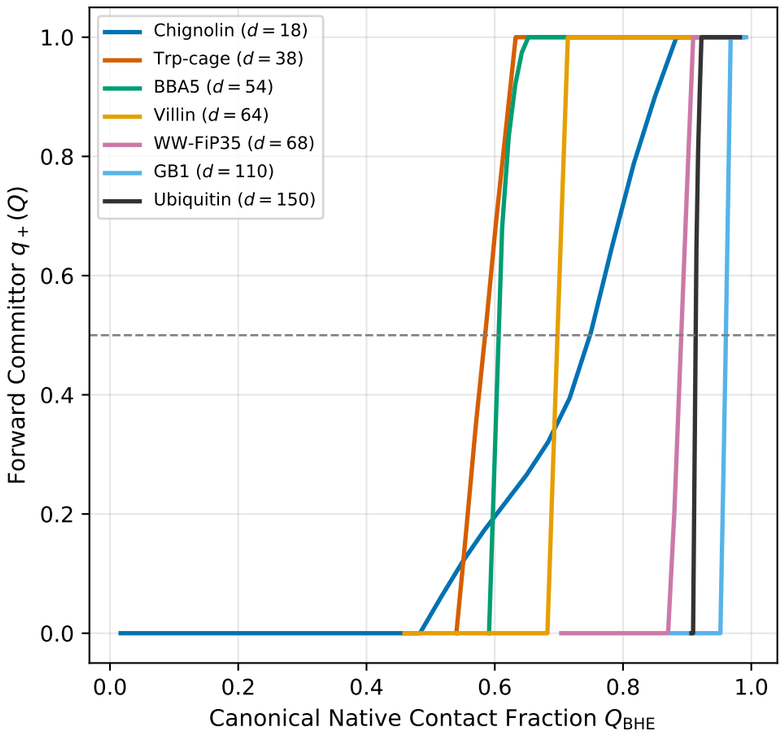
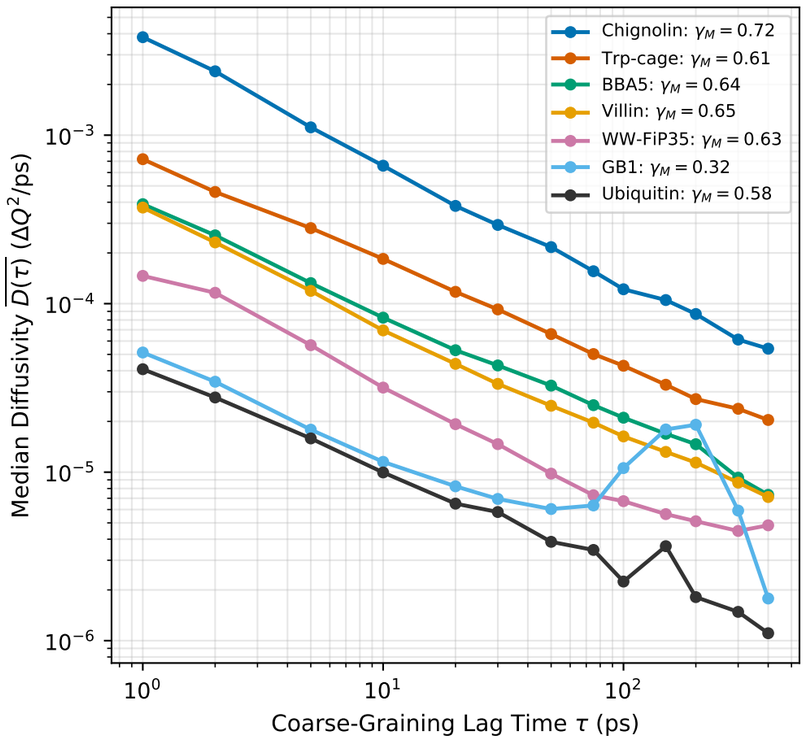
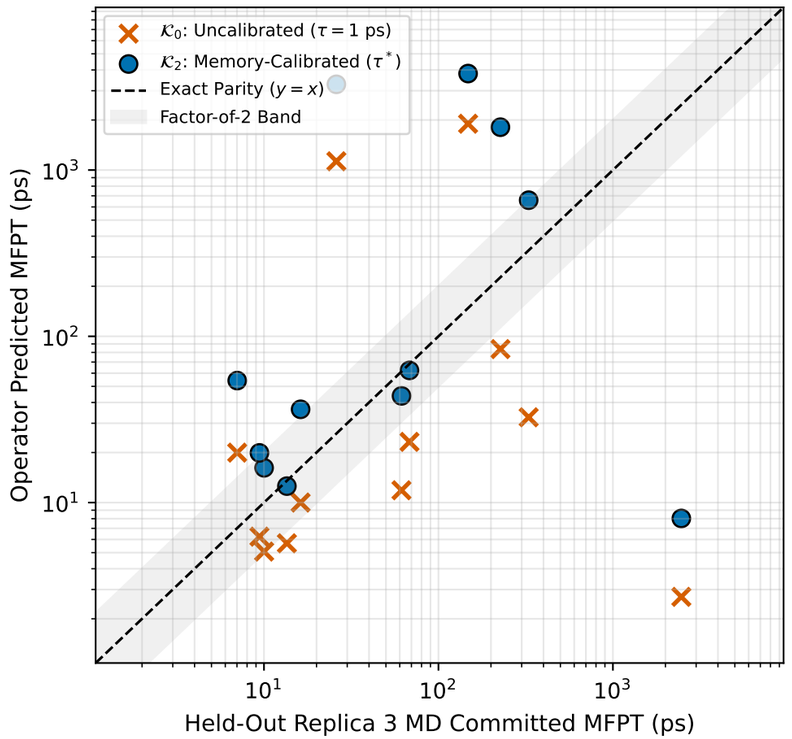
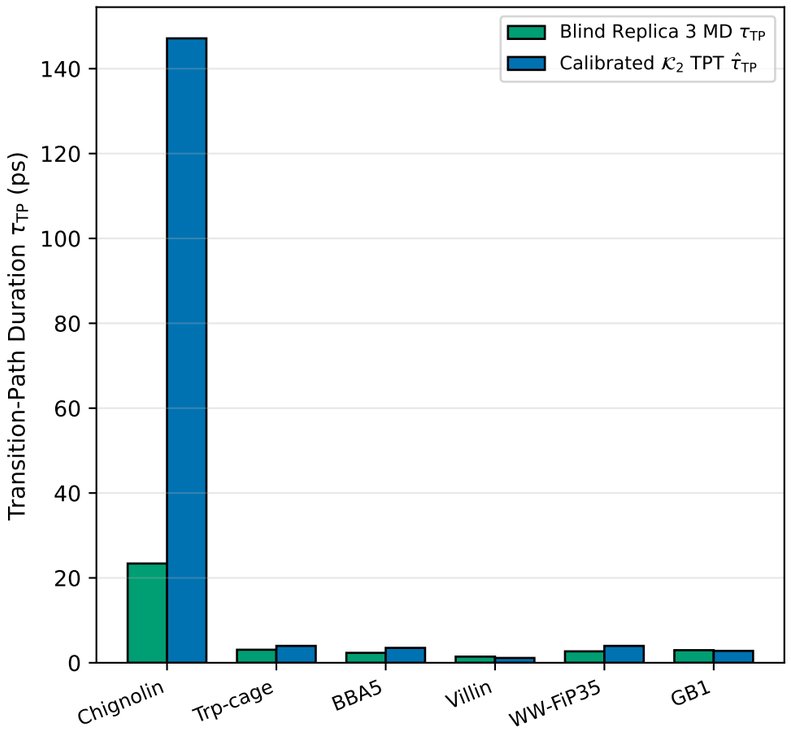
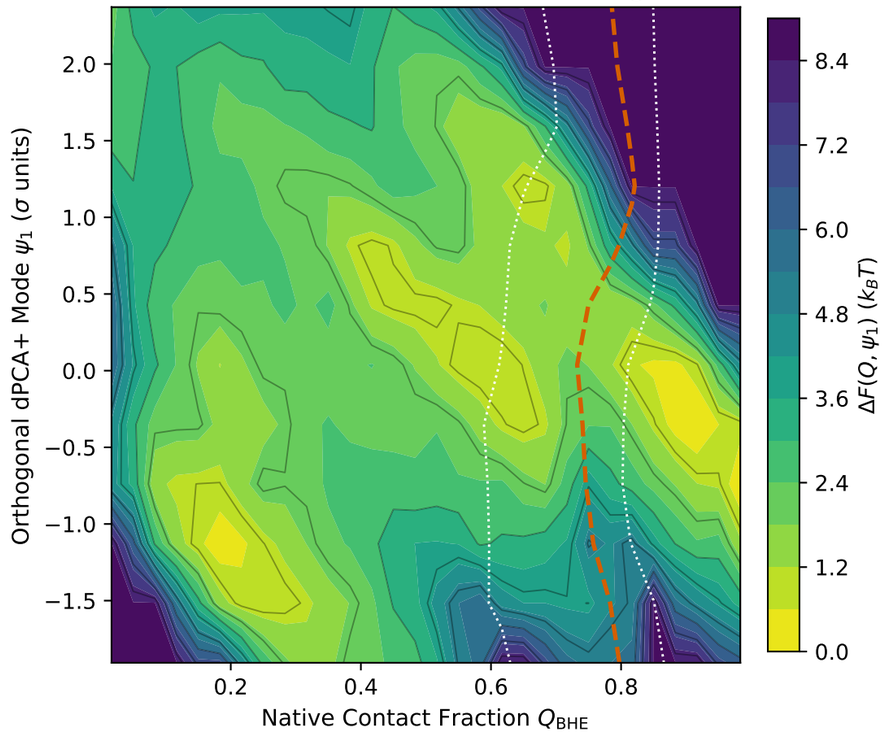
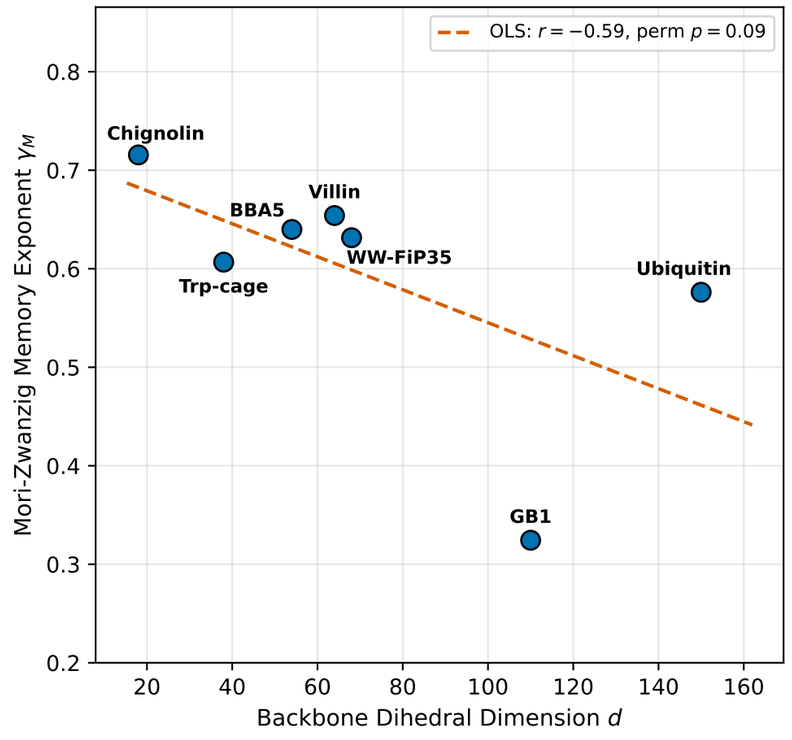

(A) Multi-Protein Committors & Separatrices
Chignolin ($d = 18$)
Trp-cage ($d = 38$)
BBA5 ($d = 54$)
Villin ($d = 64$)
WW-FiP35 ($d = 68$)
GB1 ($d = 110$)
Ubiquitin ($d = 150$)
Forward Committor $q_+(Q)$
Canonical Native Contact Fraction Q_{BHE}


(B) Mori-Zwanzig Sub-Diffusive Memory Decay
Chignolin: $\gamma_M = 0.72$
Trp-cage: $\gamma_M = 0.61$
BBA5: $\gamma_M = 0.64$
Villin: $\gamma_M = 0.65$
WW-FiP35: $\gamma_M = 0.63$
GB1: $\gamma_M = 0.32$
Ubiquitin: $\gamma_M = 0.58$
Median Diffusivity $\overline{D(\tau)}$ $(\Delta Q^2/\mathrm{ps})$
Coarse-Graining Lag Time τ (ps)


(C) Blind Test MFPT Parity Across Hierarchy
$\mathcal{K}_0$: Uncalibrated ($\tau = 1$ ps)
$\mathcal{K}_2$: Memory-Calibrated (τ^*)
Exact Parity ($y = x$)
Factor-of-2 Band
Operator Predicted MFPT (ps)
Held-Out Replica 3 MD Committed MFPT (ps)


(D) Hummer Transition-Path Duration
Blind Replica 3 MD τ_{TP}
Calibrated $\mathcal{K}_2$ TPT $\hat{\tau}_{\mathrm{TP}}$
Transition-Path Duration τ_{TP} (ps)
Chignolin
Trp-cage
BBA5
Villin
WW-FiP35
GB1


(E) Empirical 2D PMF & $q_+ = 0.5$ (Chignolin)
Orthogonal dPCA+ Mode ψ_1 (σ units)
Native Contact Fraction Q_{BHE}
$\Delta F(Q, \psi_1)$ $(k_B T)$


(F) Cross-Protein Memory Exponent Scaling
OLS: $r = -0.59$, perm $p = 0.09$
Chignolin
BBA5
Villin
WW-FiP35
Trp-cage
Ubiquitin
GB1
Mori-Zwanzig Memory Exponent γ_M
Backbone Dihedral Dimension d